

Mirriam Chisunta Chikwamu

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EDUCATION

Brown University

Master of Science (ScM), Biostatistics

August 2026 - present

Relevant Coursework: Generalized Linear Models, Statistical Programming, Theoretical Probability

Whitman College

Bachelor of Arts, Mathematics-Statistics; Minor: Economics

Dec 2024

EXPERIENCE

Symbolic AI Researcher

July 2025 – February 2026

Oak Ridge Institute for Science and Education | Navy Postgraduate School

Monterey, CA

- Engineered formal logic representations by translating natural language into SUO-KIF axioms for SUMO ontology, with approved definitions integrated into SigmaKEE's live production instance
- Built Java-based proof translation system converting ATP outputs into Basic English, enabling non-technical military stakeholders to interpret formally verified reasoning
- Evaluated 400+ sentences for semantic validity to create a labeled dataset for validating logical inference pipelines

Logic & Computation Teacher

2025 – 2026

Great Hearts Trivium Preparatory Academy

Goodyear, AZ

- Taught a two-semester senior course tracing formal logic from Aristotle to Python; rebuilt the course mid-year and designed a senior honors elective
- Mentored students through milestone-based final projects (proposal, checkpoint, final code, presentation)

Statistician Intern

Aug 2024 – Sept 2024

Economic and Financial Statistics Division, Zambia Statistical Agency

Lusaka, Zambia

- Flagged 100+ pricing anomalies across 1,000+ data points by cross-referencing historical baselines, preventing corrupted data from affecting Zambia's August 2024 national economic indicators
- Redesigned legacy digital forms using SQL/Access principles, increasing survey throughput from 60 to 200 per day (3.3x) and completing a 4-week digitization project in 1 week

Sports Analytics Fellow

June 2024 – July 2024

Department of Statistics & Data Science, Carnegie Mellon University

Pittsburgh, PA

- Developed Weighted Mean Change framework using PCA-weighted z-scores to create a combined consistency metric from EVG, EVA, and shooting percentage, validating year-to-year stability ($R^2 = 0.18-0.34$) across 200+ NHL forwards over 8 seasons in R

Rosetta Commons Research Intern, Khare Lab

June 2023 – Aug 2023

Institute for Quantitative Biomedicine, Rutgers University

New Brunswick, NJ

- Performed computational analysis of 3C protease using MMseq2 for homology search across viral protein databases (~253M entries in UniProt), identifying catalytic residue sequence motifs through machine learning-based pattern recognition
- Identified 5 promising enzyme variants for experimental validation, confirming catalytic mechanisms from the lab's preprint and enabling the wet lab team to prioritize therapeutic design candidates

PROJECTS

Senior Thesis: Predicting NHL Forward Successes Using Random Forests

Sep 2024 – Dec 2024

math.whitman.edu/SeniorProjects/2024/

- Built Random Forest models achieving 0.90 R^2 to predict NHL forward salary cap, contract length, and performance consistency, outperforming traditional ARIMA (0.60 R^2) by integrating Tango-adjusted metrics and custom variance calculations
- Designed interactive RShiny dashboard enabling stakeholders to compare algorithmic vs. traditional model predictions with dynamic visualizations

Digital Heritage Archive (Hackathon)

April 2024

github.com/ChichiChikwa/Digital_Heritage_Archive

- Fine-tuned Gemini Vision Pro 1.2 for image-based cultural heritage destination recognition, winning Whitman College's inaugural hackathon

SKILLS

Programming Languages: Python, R, SQL, Java, C++, JavaScript

ML/AI Frameworks: TensorFlow, PyTorch, Scikit-learn, Random Forests, CNNs

Knowledge Representation: SUMO, SUO-KIF, Automated Theorem Proving (E-Prover, Vampire), SigmaKEE

Data Science & Visualization: NumPy, pandas, Seaborn, Matplotlib, ggplot2, RShiny

Statistical Methods: Regression (Linear, Logistic, LASSO, Ridge), Clustering (k-Means, DBSCAN, Hierarchical), PCA, A/B Testing, ANOVA

ACTIVITIES & LEADERSHIP

International Association of Black Actuaries

Aug 2023 – May 2024

Scholarship Recipient

Whitman African Students Association

Aug 2023 – May 2024

Events Coordinator

Chichasa NGO

Oct 2019 – Present

Co-Founder • Youth initiative in Zambia

HERZ Non-Governmental Organization

Jan 2021 – Present

Social Media Contributor